

Protein 14-3-3 Influences the Response of the Cardiac Sodium Channel Na_v1.5 to Antiarrhythmic Drugs[§]

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ABSTRACT

The cardiac sodium channel Na_v1.5 is a key contributor to the cardiac action potential, and dysregulations in Na_v1.5 can lead to cardiac arrhythmias. Na_v1.5 is a target of numerous antiarrhythmic drugs (AADs). Previous studies identified the protein 14-3-3 as a regulator of Na_v1.5 biophysical coupling. Inhibition of 14-3-3 can remove the Na_v1.5 functional coupling and has been shown to inhibit the dominant-negative effect of Brugada syndrome mutations. However, it is unknown whether the coupling regulation is involved with AADs' modulation of Na_v1.5. Indeed, AADs could reveal important structural and functional information about Na_v1.5 coupling. Here, we investigated the modulation of Na_v1.5 by four classic AADs, quinidine, lidocaine, mexiletine, and flecainide, in the presence of 14-3-3 inhibition. The experiments were carried out by high-throughput patch-clamp experiments in an HEK293 Na_v1.5 stable cell line. We found that 14-3-3 inhibition can enhance acute block by quinidine, whereas the block by other drugs was not affected. We also saw changes in the use- and dose-dependency of quinidine, lidocaine, and mexiletine when

inhibiting 14-3-3. Inhibiting 14-3-3 also shifted the channel activation toward hyperpolarized voltages in the presence of the four drugs studied and slowed the recovery of inactivation in the presence of quinidine. Our results demonstrated that the protein 14-3-3 and Na_v1.5 coupling could impact the effects of AADs. Therefore, 14-3-3 and Na_v1.5 coupling are new mechanisms to consider in the development of drugs targeting Na_v1.5.

SIGNIFICANCE STATEMENT

The cardiac sodium channel Na_v1.5 is a target of commonly used antiarrhythmic drugs, and Na_v1.5 function is regulated by the protein 14-3-3. The present study demonstrated that the regulation of Na_v1.5 by 14-3-3 influences Na_v1.5's response to antiarrhythmic drugs. This study provides detailed information about how 14-3-3 differentially regulated Na_v1.5 functions under the influence of different drug subtypes. These findings will guide future molecular studies investigating Na_v1.5 and antiarrhythmic drugs outcomes.

Introduction

Sudden cardiac death (SCD) associated with cardiac arrhythmia is the most common cause of death and remains a public health issue. SCD accounts for 15%–20% of all death worldwide, and in the United States alone, there are about 180,000–300,000 SCD cases annually (Srinivasan and Schilling, 2018; Kim et al., 2021; Tan et al., 2022). Treatments for cardiac arrhythmias include cardiac ablation, implantable cardioverter-defibrillators, and antiarrhythmic drugs (AADs) (Fu, 2015). AADs can treat cardiac arrhythmic conditions without surgical procedures and can be used as a preventive treatment. Therefore, AADs have great significance in clinical practice. Although the efficiency and safety of AADs have been tested by numerous experiments and clinical trials, not all aspects of the AADs' functions on their targets are fully

understood, and the missing information can be essential in reducing side effects, improving efficacy, developing new drugs, and gaining knowledge of the target proteins.

The cardiac sodium channel (Na_v1.5) is the predominant voltage-gated sodium channel in the human heart. It generates the cardiac sodium current (I_{Na}) that initiates the cardiac action potential, and dysregulation of I_{Na} can lead to arrhythmic phenotypes (Fonseca and Vaz da Silva, 2018). Na_v1.5 is tightly regulated by interacting protein complexes. (Gavillet et al., 2006; Abriel, 2010) Dysfunction in Na_v1.5 regulatory proteins can also lead to cardiac arrhythmias (Adsit et al., 2013; Xiong et al., 2022). As for many cardiac ion channels, Na_v1.5 is an essential target of AADs. AADs are categorized into five classes according to the Vaughan-Williams classification system. Class I AADs are sodium channel blockers. They block the sodium current to decrease the action potential's amplitude and slow conduction velocity, thus interrupting or preventing reentrant arrhythmias (Pott et al., 2010). Considering that Na_v1.5 interacting proteins can regulate the channel function, it is highly possible that certain Na_v1.5 regulatory proteins could affect the channel's response to AADs, thus influencing the AAD's outcome or side effects. However, there are limited studies that

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ABBREVIATIONS: AAD, antiarrhythmic drug; D, domain; difopein, dimeric fourteen-three-three peptide inhibitor; I/V, current/voltage relationship; I_{Na}, cardiac sodium current; Na_v1.5, voltage-gated cardiac sodium channel; SCD, sudden cardiac death; SSI, steady-state inactivation.

have investigated how the different $\text{Na}_v1.5$ regulatory proteins influence the action of AADs on $\text{Na}_v1.5$.

14-3-3 is one of $\text{Na}_v1.5$'s interacting proteins. It prevalently exists in all eukaryotic cells and regulates $\text{Na}_v1.5$'s function in the heart (Allouis et al., 2006). Notably, other than the direct modulation of $\text{Na}_v1.5$ biophysical function, 14-3-3 was found to regulate $\text{Na}_v1.5$ coupling, a novel mechanism that can contribute to arrhythmic phenotypes (Clatot et al., 2018; Zheng et al., 2021). Studies also suggested that modulating the action of 14-3-3 on $\text{Na}_v1.5$ could potentially bring therapeutic effects. This was also observed in other voltage-gated sodium channel isoforms related to neuronal disorders, therefore suggesting that 14-3-3 has a conserved role in regulating voltage-gated sodium channels (Ruhlmann et al., 2020). However, the underlying mechanism of how 14-3-3 regulates $\text{Na}_v1.5$ coupling is not fully understood, which limits its potential use as a therapeutic strategy. Additionally, it is unknown if the biophysical coupling regulated by 14-3-3 is involved in AADs' action on $\text{Na}_v1.5$. Thus, in this paper, we proposed to study the influence of protein 14-3-3 on class I AADs' effects on the cardiac sodium channel. Cardiac sodium currents were studied by high-throughput automated patch-clamping using an HEK293 $\text{Na}_v1.5$ stable cell line. The specific 14-3-3 inhibitor dimeric fourteen-three-three peptide inhibitor (difopein) was used to determine if 14-3-3 and the sodium channel coupling impact the effects of AADs on $\text{Na}_v1.5$.

Materials and Methods

Materials. Quinidine (Q3625), flecainide (F6777), lidocaine (PHR1257), and mexiletine (M2727) were obtained from Sigma (St. Louis, MO). Drugs were diluted into a 100-mM stock solution in water, methanol, or ethanol according to the manufacturer's instruction. Working concentrations of 20 (quinidine), 30 (flecainide), and 100 μM (lidocaine and

mexiletine) were freshly prepared in the electrophysiology extracellular solution.

Electroporation. The 14-3-3 inhibitor difopein (pEYFP-C1-difopein) (Masters and Fu, 2001; Clatot et al., 2017) and negative control (pcDNA3.1) were transiently transfected into HEK293 cells by electroporation using ATx from MaxCyte (Gaithersburg, MD). HEK293 cells were plated 24 hours before electroporation and harvested on the day of electroporation at 70%–80% confluency. Cells were collected by centrifugation at 1.1 rpm for 5 minutes followed by washing with electroporation buffer (MaxCyte, Gaithersburg, MD). The cells were then resuspended to the concentration of 10^6 cells/ μl and mixed with the plasmid of interest at the concentration of 200 ng/ μl . The cells were electroporated and then incubated at 37°C for 20 minutes. Following incubation, the cells were transferred to 10-cm dishes and grown for 48 hours until patch-clamp experiments. Difopein expression level was confirmed before each patch-clamp experiment by observing the YFP signal under a fluorescent microscope.

Cell Culture and Expression of $\text{Na}_v1.5$. HEK293 $\text{Na}_v1.5$ stable cell line was maintained in Dulbecco's modified Eagle's medium (Gibco, Waltham, MA) supplemented with 10% FBS (Gibco, Waltham, MA) until used for transfection and electrophysiological recordings; 400 $\mu\text{g}/\text{ml}$ G418 (Sigma, St. Louis, MO) was added to the medium for selection of $\text{Na}_v1.5$ -HEK293 cells. All cells were grown at 37°C in 5% CO_2 .

Electrophysiology Measurement. Sodium currents were measured by patch-clamping in the whole-cell configuration 48 hours after transfection. The recordings were obtained using the high-throughput automated patch-clamp system Syncroptach 384i (Nanion, Germany). The patch-clamp experiments were conducted with high-throughput automated patch-clamp system Syncropatch 384i (Nanion, Germany). Compared with the traditional manual patch-clamp technology, the automated patch-clamp greatly increased the efficiency while eliminating operators' bias. The high number of cells being recorded all at the same time greatly reduced variability compared with manual patch-clamp and is particularly beneficial when studying pharmacological effects since it allows for all the cells to be exposed to the drugs for the exact same time. But most importantly, in our hands, the automated patch-clamp yields a highly consistent

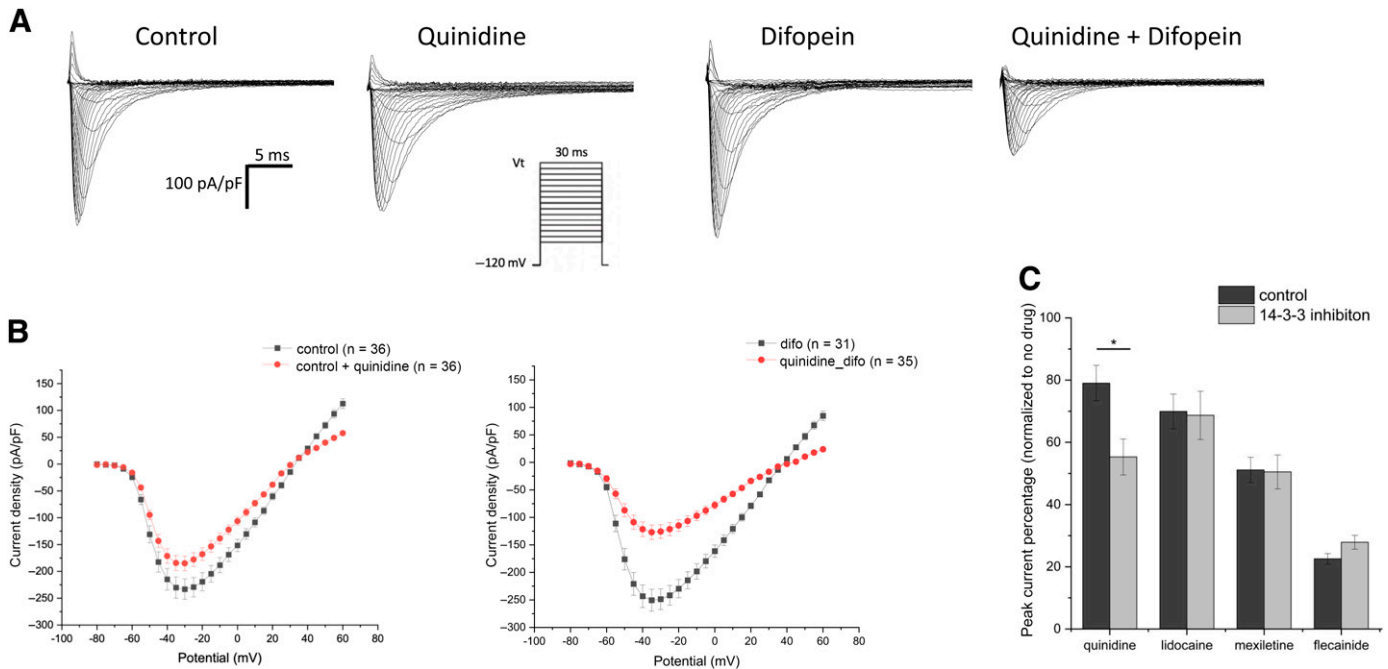


Fig. 1. 14-3-3 inhibition increased $\text{Na}_v1.5$ block by quinidine. The sodium currents were recorded in HEK293 cells by high-throughput automated patch-clamp in whole-cell configuration. 14-3-3 was inhibited by difopein (difo). (A) Representative current traces for each condition. (B) Current-voltage (I/V) relationship. Voltage protocol is included. (C) Quantification of the sodium current block by all four AADs. Statistical analysis was done by two-tailed paired t test.

current profile compared with the manual patch-clamp in parallel (data not shown), illustrating that the data fully compare with manual patch-clamp. This observation is similar to the previous recordings of Nav1.7 channel with the high-throughput patch-clamp system previously reported (Li et al., 2017). The intracellular solution contained (in mM) 10 NaCl, 110 CsF, 10 CsCl, 10 EGTA, and 10 HEPES, pH 7.4. The extracellular solution contained (in mM) 30 NaCl, 4 KCl, 2 CaCl₂, 1 MgCl₂, 5 glucose, 10 HEPES, and 110 NMDG, pH 7.4. Since sodium current density was high, low sodium concentration was used in the external solution, and NMDG was used to balance the osmolarity. The current-voltage relationships of the sodium currents were recorded by holding the resting membrane potential at -120 mV and stepping from -80 mV to +60 mV in a 5-mV interval (each step held for 30 milliseconds). For activation, the conductance/voltage relationship curve was determined by fitting the linear part before the peak of the current-voltage curve with a Boltzmann function:

$$G(V) = G_{\max} / (1 + e^{-(V - V_{\text{half}})/k}) \quad (1)$$

Recovery from inactivation was recorded with a two-pulse protocol. The prepulse and the test-pulse duration are 30 milliseconds, stepping from -30 mV to -120 mV. The interval between the two pulses ranges from 1.8 milliseconds to 250 milliseconds. Currents of the recovery from inactivation were fitted to the following equation:

$$I_{\text{test}} / I_{\text{pre-pulse}} = A_{\text{rec}} (1 - e^{-t/\tau_{\text{rec}}}) + i_{\text{rec}} \quad (2)$$

The steady-state inactivation (SSI) was studied with a 500-millisecond prepulse ranging from -140 mV to -30 mV, followed by a 30-millisecond test pulse stepping from -120 mV to -30 mV. The currents for the steady-state inactivation were fit to a Boltzmann distribution using the following equation:

$$I/I_{\max} = (1 + e^{(V - V_{1/2})/k_v})^{-1} \quad (3)$$

The dose-dependency was studied with a single pulse from -120 to -10 mV for 30 milliseconds. Cells harvested from the same dish were separated into different groups by the container of the automated patch-clamp system; the system then applied varied concentrations of the drugs of interest to different groups, and the sodium currents were recorded simultaneously from each group. The current/voltage relationship (I/V) curves were recorded to confirm voltage control. Dose-dependency curves were fitted by the dose-response fitting with the following equation:

$$I/I_{\max} = A1 + \frac{A2 - A1}{1 + 10^{(\log x_0 - x)/p}} \quad (4)$$

Use-dependency was studied with train pulses at 1, 5, and 10 Hz. The holding potential was -120 mV, and the test pulses were -10 mV. The use-dependency curves were fitted with an exponential function.

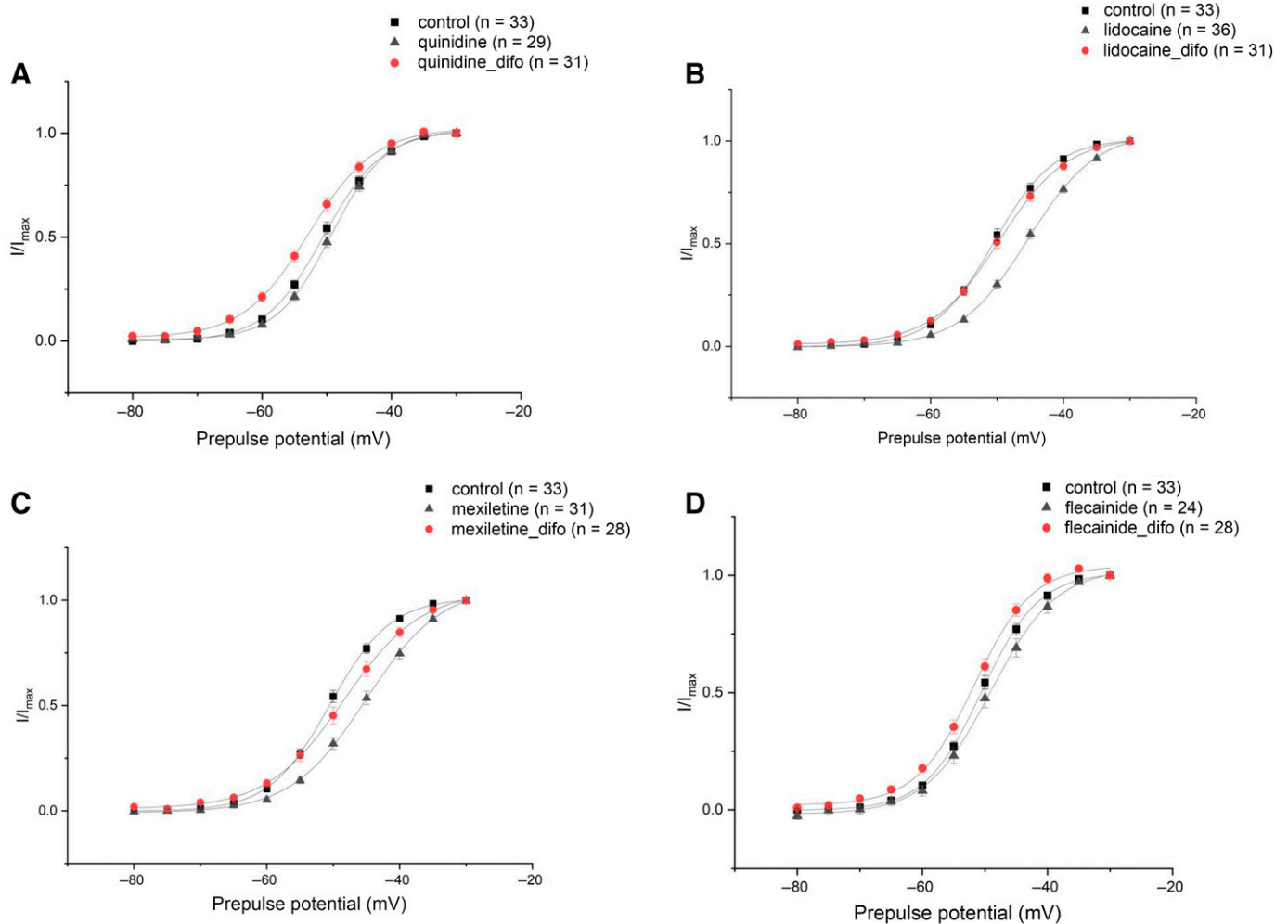


Fig. 2. 14-3-3 inhibition shifted the conductance curves under the influence of AADs. The voltage-dependent conductance curves for quinidine (A), lidocaine (B), mexiletine (C), and flecainide (D) were derived from the I/V curves as described in *Materials and Methods*. Quinidine (A) and flecainide (D) did not exert significant effect on the activation, whereas lidocaine (B) and mexiletine (C) shifted the activation curves toward depolarizing voltages. Compared with the drug-only effects, 14-3-3 inhibition shifted all the activation curves toward hyperpolarizing voltages. difo, difenyl; $I/I_{\max}$, the current recorded at indicated voltage level over the maximum current.

All fitting curves were generated with Origin 10.1.1 software (OriginLab Corporation, Northampton, MA). The leak subtraction protocol was used for all recording protocols. Cells with I_{Na} amplitude over 700 pA/pF or with incomplete space-clamp were excluded from data analysis (Montnach et al., 2021).

Statistical Analysis. Statistical analyses were performed using the standard statistical package in origin 10.1.1 (OriginLab Corporation, Northampton, MA). The Student's t test was performed at a significance level of $P = 0.05$ for a single comparison. One-way ANOVA with post hoc Tukey tests were performed for comparison among three or more groups, and P values less than 0.05 were considered statistically significant. Results were presented as mean $\pm$ S.E.M.

Results

Protein 14-3-3 is a family of highly conserved regulatory proteins. They were shown to bind with $Na_v1.5$ on the channel's intracellular loop and regulate $Na_v1.5$ function. 14-3-3 protein-protein interaction can be inhibited by a specific 14-3-3 inhibitor, difopein. As indicated by the name, difopein con-

tains two repeats of the R18 peptide, which occupied with high affinity the 14-3-3 groove supposed to bind to its target proteins (Wang et al., 1999). The dimeric version of R18 (difopein) further improves the efficiency of 14-3-3-inhibitor binding, thereby disrupting 14-3-3 interacting with other proteins (Stevens et al., 2018). Previous studies found that inhibiting 14-3-3 could remove the functional coupling of the voltage-gated sodium channel (Clatot et al., 2017). Uncoupling the wild-type $Na_v1.5$ with Brugada syndrome mutations or a splice variant associated with heart failure can restore the sodium current impaired by the mutation (Clatot et al., 2018; Zheng et al., 2021). Thus, these previous findings suggested that 14-3-3 inhibitors could be a potential therapeutic strategy for arrhythmias linked to sodium channel dysfunction. However, since the molecular mechanism of the regulation of $Na_v1.5$ by 14-3-3 is still not completely understood and, importantly, since 14-3-3 regulates the function of numerous proteins in the human body, additional investigations are needed to eventually develop therapeutic strategies targeting 14-3-3 for cardiac arrhythmias.

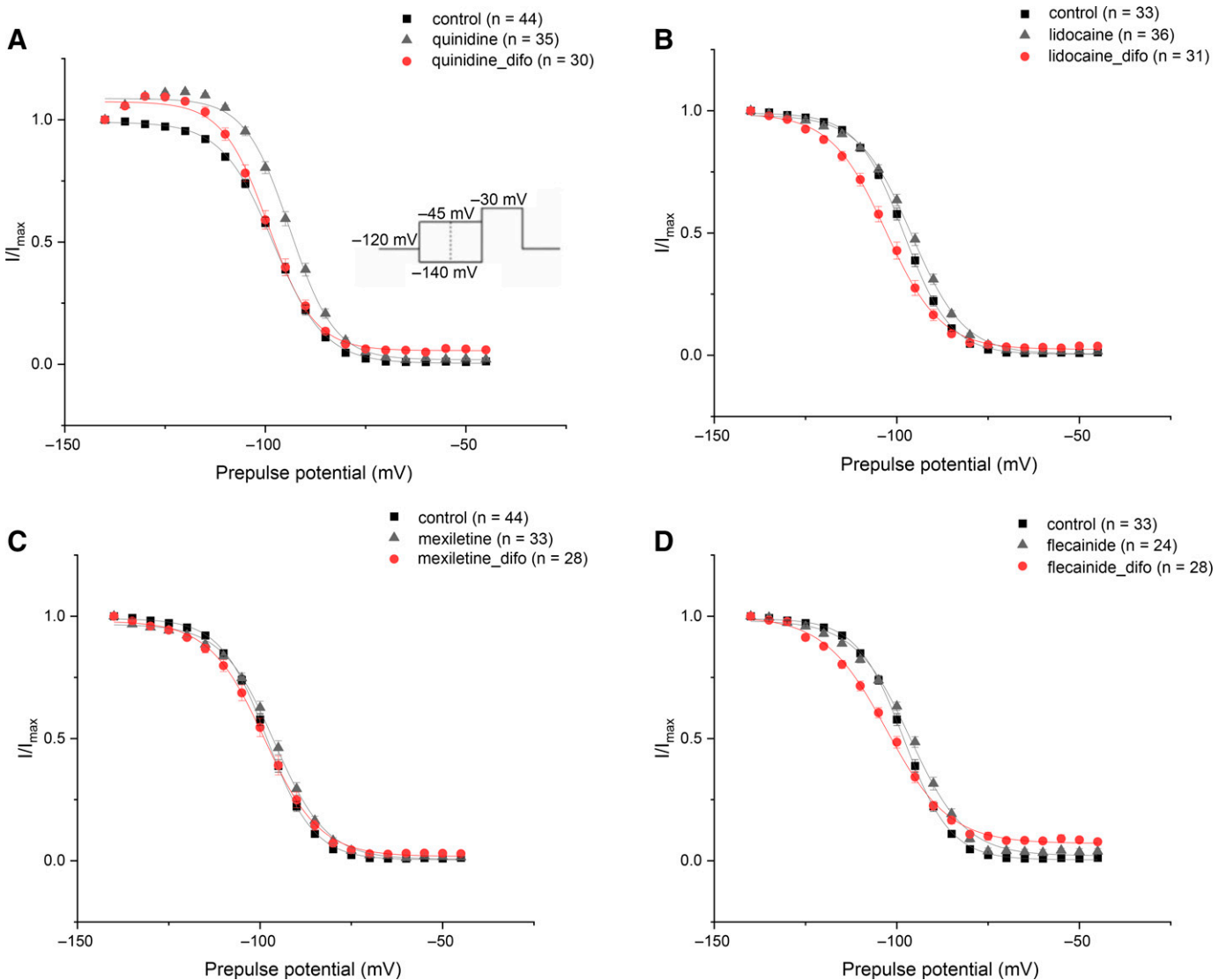


Fig. 3. 14-3-3 impact on SSI curves of the sodium current under the influence of AADs. The sodium currents were recorded from HEK293 cells in whole-cell configuration. SSI curves for quinidine (A), lidocaine (B), mexiletine (C), and flecainide (D) and the voltage protocol are shown in the figure. 14-3-3 inhibition removed the quinidine effect on SSI and shifted the SSI of lidocaine (B) and flecainide (D) toward hyperpolarizing voltages. difo, difopein; I/I_{max} , the current recorded at indicated voltage level over the maximum current.

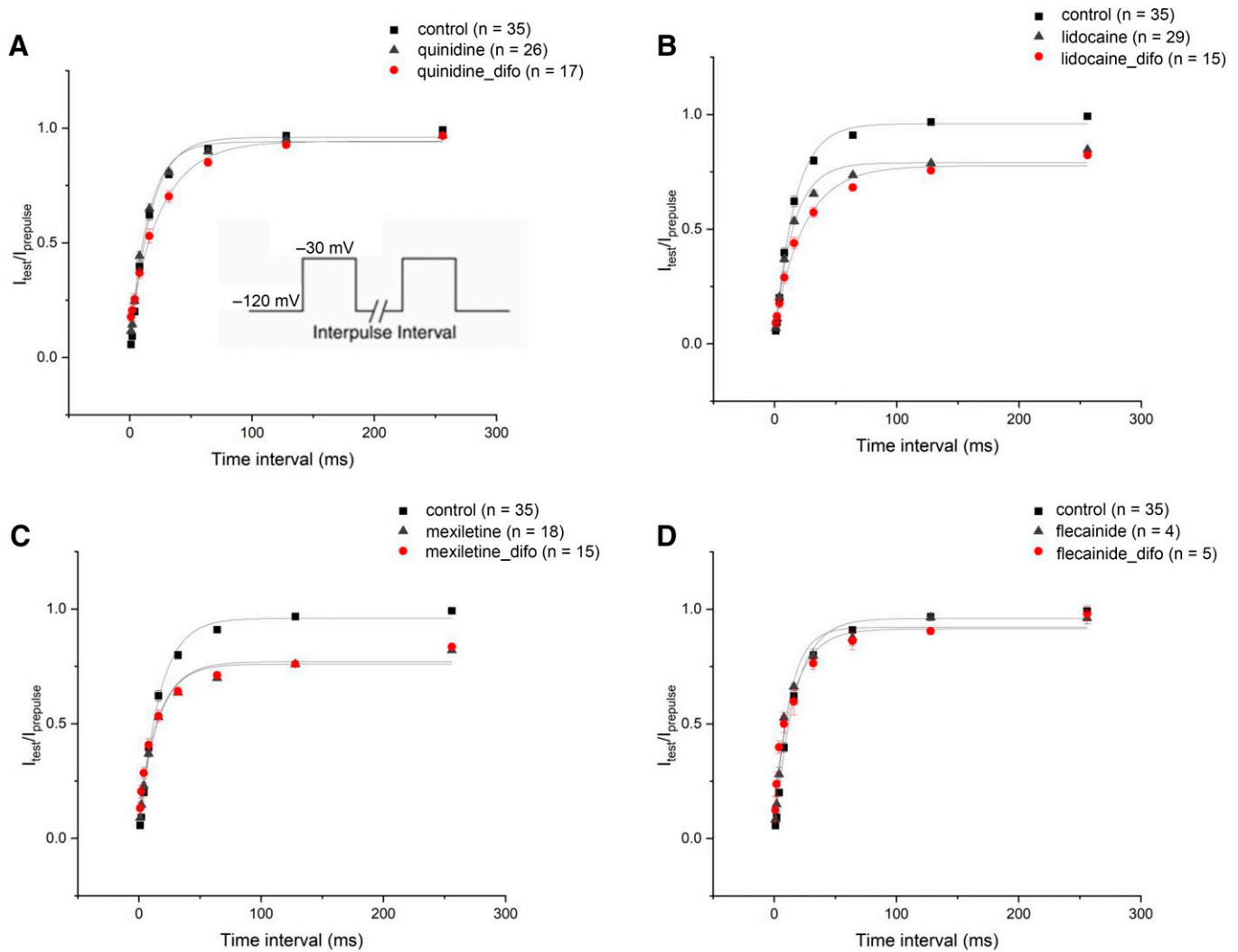


Fig. 4. Impact of 14-3-3 on sodium channel's recovery from inactivation under the influence of AADs. The sodium currents were recorded from HEK293 cells in whole-cell configuration. Recovery curves for quinidine (A), lidocaine (B), mexiletine (C), and flecainide (D) and the voltage protocol are shown in the figure. Lidocaine (B) and mexiletine (C) slowed the channel recovery from inactivation without the involvement of 14-3-3, whereas 14-3-3 inhibition further slowed the channel recovery for the quinidine (A) and flecainide (D). difo, difopein; $I_{\text{test}}/I_{\text{prepulse}}$, the current recorded at the test pulse over current recorded at the prepulse.

14-3-3 Inhibition Aggravated $\text{Na}_v1.5$ Block by Quinidine.

To account for the impact of electroporation transfection effects, HEK293 cells transfected with 14-3-3 inhibitor difopein were compared with those transfected with pcDNA3.1. For simplicity, the groups transfected with control plasmid (pcDNA3.1) and recorded in the absence of AADs were labeled "control," the groups transfected with control plasmid and treated with the AAD were labeled by the AAD's name (i.e., "quinidine"), and the groups transfected with difopein and treated with the AAD were labeled by the AAD's name_difo (i.e., 'quinidine_difo'). Cardiac sodium currents with and without 14-3-3 inhibition was recorded from HEK293 $\text{Na}_v1.5$ stable cell line in the presence of quinidine (class Ia), lidocaine (class Ib), mexiletine (class Ib), and flecainide (class Ic). The I/V curves were plotted to show the voltage-dependency in the presence of AADs and 14-3-3 modulation. As expected, all four drugs blocked sodium currents. 14-3-3 inhibition and AADs' block did not change the current-voltage relationship of the sodium channel. However, 14-3-3 inhibition significantly aggravated the block by quinidine, whereas

the current reductions for the other drugs remained unchanged in the presence of 14-3-3 inhibition. Fig. 1A shows representative examples of the sodium currents reduction caused by quinidine with and without 14-3-3 inhibition. The I/V relationships are illustrated in Fig. 1B. The peak current was reduced from -233.14 ± 18.96 pA/pF ($n = 36$) to -185.10 ± 13.66 pA/pF ($n = 33$) in the presence of quinidine in the control group and from -248.68 ± 18.97 pA/pF ($n = 31$) to -125.63 ± 12.47 pA/pF ($n = 35$) in the difopein (14-3-3 inhibitor) group. However, 14-3-3 inhibition did not modify the block in sodium currents by lidocaine, mexiletine, and flecainide (Fig. 1C). I/V curves for lidocaine, mexiletine, and flecainide are shown in Supplemental Fig. 1. We also obtained $\text{Na}_v1.5$ conductance curves from the current-voltage relationship. We found that when the AADS were applied in the presence of 14-3-3 inhibition, all the activation curves were shifted toward hyperpolarizing voltages compared with the response to AADs alone (Fig. 2). Interestingly, with 14-3-3 inhibitor difopein, the resulting G/V curves for lidocaine plus difopein and mexiletine plus difopein behave

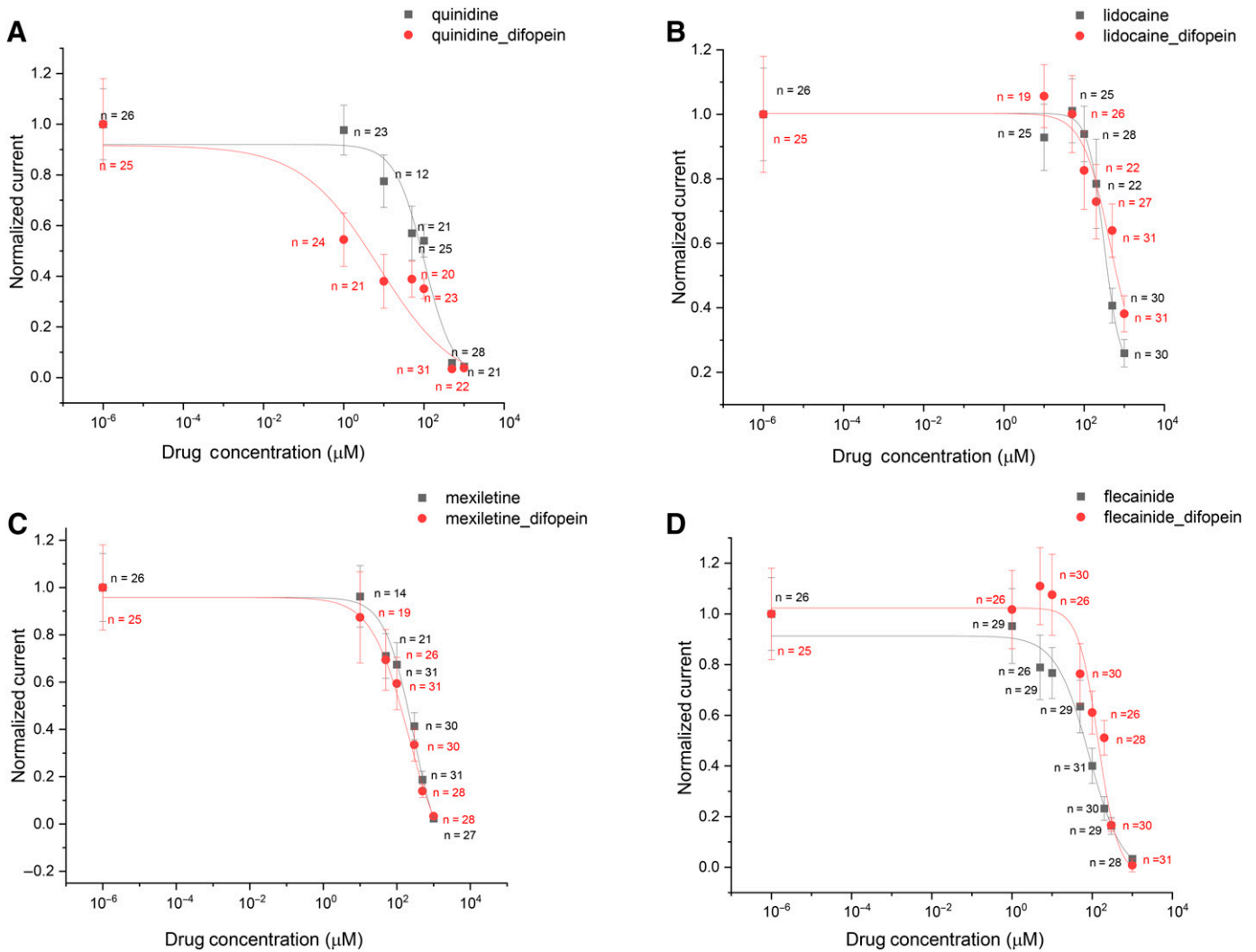


Fig. 5. 14-3-3 impacts quinidine's dose-dependent block. The dose-dependency of quinidine (A), lidocaine (B), mexiletine (C), and flecainide (D) are shown in the figure. 14-3-3 inhibition lowered the IC_{50} of quinidine (A).

similarly to the control groups in which the corresponding drugs were absent (Fig. 2, B and C). This suggests that 14-3-3 is involved in the regulation of the conductance by class Ib AADs.

14-3-3 Inhibition Altered $Na_v1.5$ Steady-State Inactivation and Recovery from Inactivation in Response to AADs.

To elucidate the influence of 14-3-3 on AADs' effects on channel kinetics, we examined $Na_v1.5$ SSI and recovery from inactivation in the presence of AADs with and without 14-3-3 inhibition. Although quinidine shifted the channel SSI toward depolarized voltages, interestingly, this shift was absent with 14-3-3 inhibition (Fig. 3A). No change in SSI was observed when lidocaine and flecainide were applied; however, when the drugs were applied in the presence of difopein, SSI was shifted toward hyperpolarized voltages (Fig. 3, B and D). For mexiletine, we saw no effect on SSI with or without difopein (Fig. 3C). We then assessed recovery from inactivation. We found that both class Ib AADs, lidocaine and mexiletine, induced a profound slowing in the recovery from inactivation; however, this slowing in the recovery was still present with 14-3-3 inhibition (Fig. 4, B and C), suggesting that 14-3-3 is not involved with this AAD's effect on I_{Na} . In the presence of quinidine, we did not see a change in

the recovery from inactivation; however, when the drug was applied in the presence of 14-3-3 inhibition, recovery from inactivation was slowed (Fig. 4A). No obvious changes were seen in recovery from inactivation for flecainide with or without 14-3-3 inhibition. (Fig. 4D) A summary of the biophysical parameters is shown in Table 1.

Impact of 14-3-3 on AADs' Dose-Dependent Block.

AADs' effects on $Na_v1.5$ depend on the concentration of drugs applied, which is referred to as the dose-dependency of the inhibition. To study whether 14-3-3 affects the AADs' dose-dependent block, we recorded sodium currents at different drug concentrations with or without 14-3-3 inhibition. The normalized sodium currents were plotted, and dose-dependency curves were generated by curve fitting. The results showed that 14-3-3 inhibition significantly changed the dose-dependency for quinidine, a class Ia antiarrhythmic drug (Fig. 5A). The calculated IC_{50} values without and with 14-3-3 inhibition were 102.1 μM and 7.4 μM , respectively. This suggests that a much smaller quinidine concentration is needed to have the same level of sodium currents blocked. On the other hand, 14-3-3 inhibition tended to move the dose-dependent curve of flecainide in the opposite direction (Fig. 5D). The IC_{50} values for flecainide were 84.3 μM and

TABLE 1
Na_v1.5 biophysical parameters

Parameters Conditions	Activation $V_{1/2}$ (mV)	n	SSI $V_{1/2}$ (mV)	n	A_{rec}	i_{rec}	τ_{rec} (ms)	n
Control	-50.50 ± 0.11	33	-98.06 ± 0.12	44	0.96 ± 0.0018	0.98 ± 0.0085	17.47 ± 1.32	35
Quinidine	-49.32 ± 0.08	29	-93.89 ± 0.42^a	35	0.94 ± 0.0029^a	0.91 ± 0.018^a	15.05 ± 1.08	26
Quinidine_difopein	-52.77 ± 0.14^c	31	-99.21 ± 0.38^b	30	0.94 ± 0.0081^a	0.81 ± 0.024^c	28.53 ± 3.22^c	17
Lidocaine	-45.34 ± 0.14^a	36	-95.57 ± 0.24^a	46	0.79 ± 0.0044^a	0.77 ± 0.014^a	16.71 ± 1.24	29
Lidocaine_difopein	-49.68 ± 0.11^b	31	-102.96 ± 0.26^b	32	0.77 ± 0.0057^a	0.71 ± 0.013^c	25.54 ± 2.39^c	15
Mexiletine	-45.08 ± 0.19^a	31	-96.11 ± 0.27	33	0.76 ± 0.0060^a	0.70 ± 0.014^a	16.18 ± 1.35	18
Mexiletine_difopein	-48.39 ± 0.14^b	28	-98.82 ± 0.25	28	0.77 ± 0.0073^a	0.66 ± 0.027^c	19.18 ± 2.85	15
Flecainide	-49.08 ± 0.23	24	-96.36 ± 0.41	12	0.92 ± 0.023^a	0.89 ± 0.03^a	12.11 ± 5.24	4
Flecainide_difopein	-52.27 ± 0.08^b	29	-102.81 ± 0.38^c	24	0.91 ± 0.022^a	0.80 ± 0.049	20.34 ± 8.33	5

A_{rec} , recovery curve-fitting amplitude; i_{rec} , recovery curve-fitting offset; τ_{rec} , recovery time constant; $V_{1/2}$, half maximum effect voltage.

^aSignificantly different from control ($P < 0.05$).

^bSignificantly different from the corresponding drug group ($P < 0.05$).

^cSignificantly different from control and the corresponding drug group ($P < 0.05$).

141.3 μM with and without 14-3-3 inhibition, respectively. For the class Ib AADs, lidocaine and mexiletine, there were no obvious changes in the dose-dependency curves, and the changes in IC_{50} values were not statistically

significant (Fig. 5). The IC_{50} values for lidocaine were 198.6 μM and 192.0 μM and were 285.9 μM and 208.3 μM for mexiletine before and after 14-3-3 inhibition, respectively.

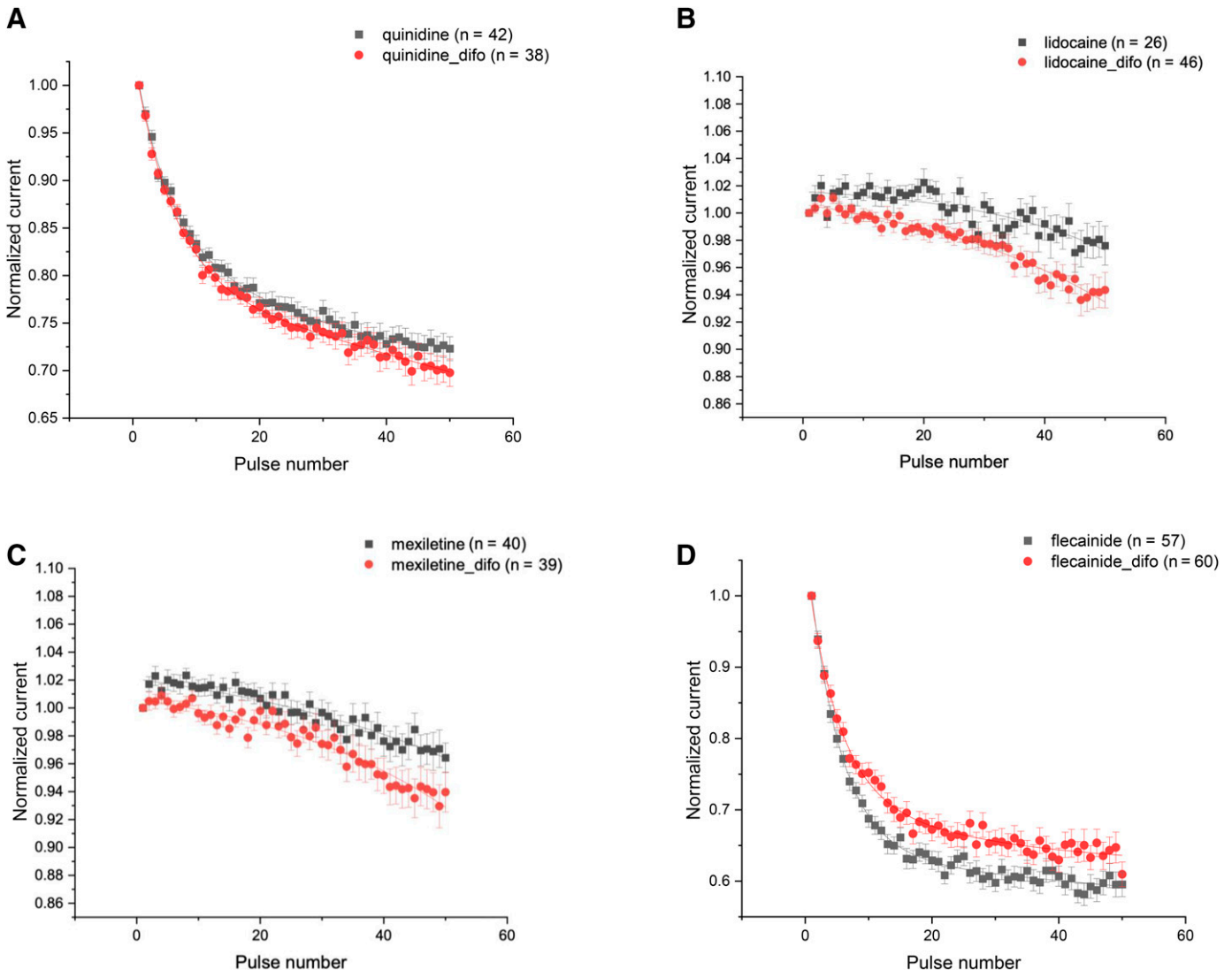


Fig. 6. Impact of 14-3-3 inhibition on the use-dependent block at 1 Hz stimulation. A train of 50 pulses was applied to the channel. The holding potential was -120 mV, and the test pulses were -30 mV for 10 milliseconds. 14-3-3 inhibition had no impact on the use-dependent block of the four drugs tested at 1 Hz. difo, difopein.

14-3-3 Altered Use-Dependency of $\text{Na}_v1.5$ on Lidocaine.

The sodium currents blocked by a drug reflect the rate of binding and releasing of the drug to or from the sodium channel. This rate can be altered by high-frequency stimulation, known as a use-dependent or frequency-dependent block. All AADs (quinidine, lidocaine, mexiletine, and flecainide) studied in this paper are known to block $\text{Na}_v1.5$ in a use-dependent manner, meaning that the channel's availability would decrease progressively with a train of stimulation (Hondeghe and Matsubara, 1988; Sunami et al., 1993; Zhu et al., 2006; Cummins, 2007). To test whether 14-3-3 influences the use-dependent block by AADs, we applied depolarization trains to the channel with varied frequencies (1, 5, and 10 Hz). At 1 Hz, which correlates with physiologic heart rate, we applied 50 pulses, and 30 pulses were applied for higher frequencies (5 and 10 Hz). All AADs exhibited use-dependent block with or without 14-3-3 inhibition (Figs. 6–8). There was no significant effect of 14-3-3 inhibition on the use-dependency of quinidine at any tested frequency (Fig. 9A). At 1 Hz, 14-3-3 inhibition increased the use-dependent inhibition of the class Ib drug lidocaine (Figs. 6B and 9B). However, at higher frequencies, 14-3-3 inhibition significantly reduced the use-dependent block by

lidocaine and mexiletine (Fig. 7, B and C; Fig. 8, B and C; Fig. 9, B and C). In the flecainide group, 14-3-3 inhibition had no effect on the use-dependent block at 1 Hz (Figs. 6D and 9D) but significantly attenuated the use-dependent block at 5 Hz (Figs. 7D and 9D). In summary, 14-3-3 inhibition changed the use-dependent block of lidocaine, mexiletine, and flecainide and exerted the most drastic effects at 5 Hz suggesting the involvement of 14-3-3 in use-dependent blocks of class Ib and class Ic AADs.

Discussion

In this study, we examined the influence of protein 14-3-3 on the cardiac sodium channel's response to four commonly used antiarrhythmic drugs. These drugs are often applied as control compounds to study the sodium channel or novel pharmaceutical agents (Hondeghe and Matsubara, 1988; Ramos and O'leary, 2004; Zhu et al., 2006; Cummins, 2007; Heath et al., 2011; Huang et al., 2013; Galleano et al., 2021; Li et al., 2021). Thus, they were selected here to study the influence of 14-3-3 on $\text{Na}_v1.5$'s pharmaceutical response. Our results demonstrated that 14-3-3 could affect the pharmacological sensitivity of $\text{Na}_v1.5$ toward clinically used drugs, providing further mechanistic

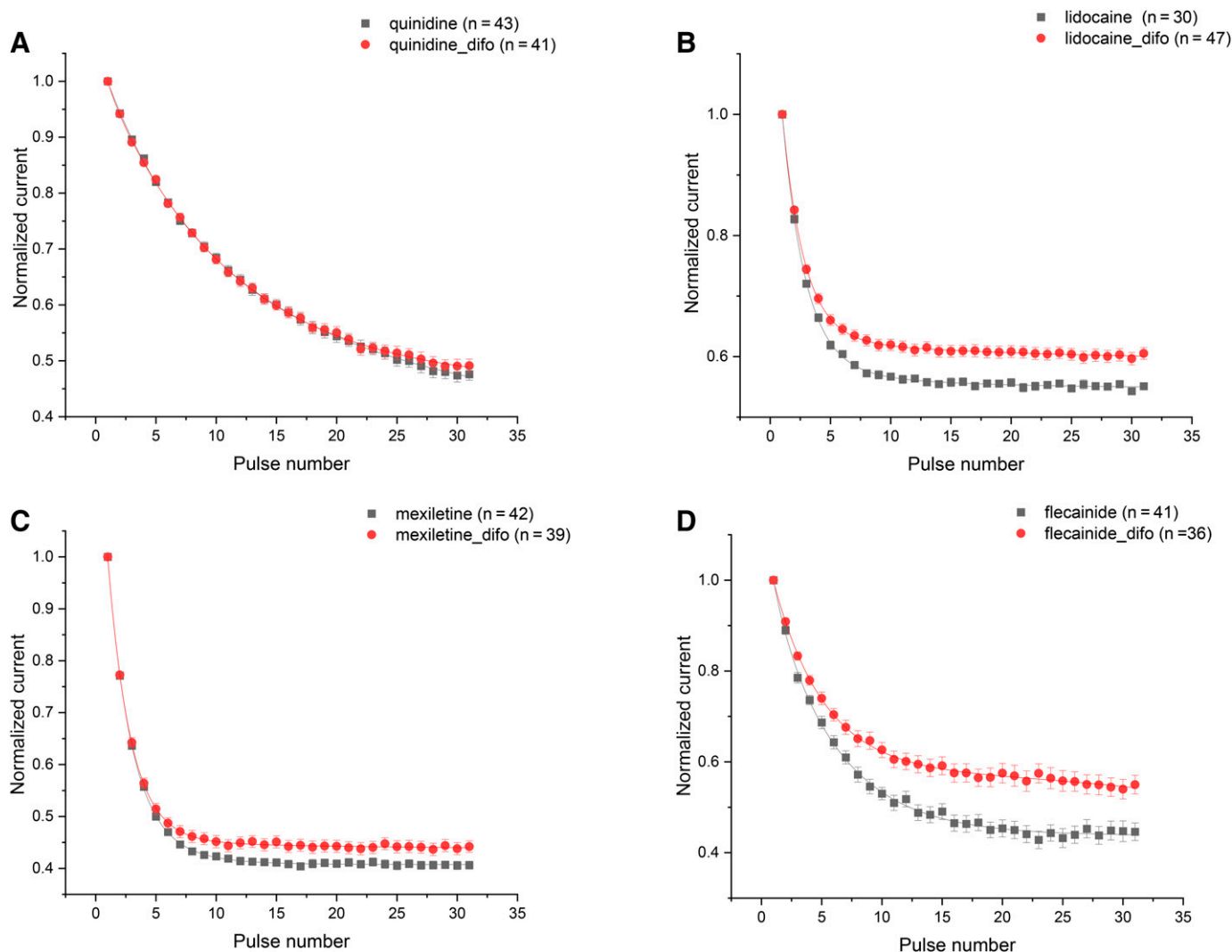


Fig. 7. Impact of 14-3-3 inhibition on the use-dependent block at 5 Hz stimulation. A train of 30 pulses was applied to the HEK293 cells. 14-3-3 inhibition diminished use-dependent block of lidocaine (B), mexiletine (C), and flecainide (D) but not quinidine (A). difo, difopein.

insights into the action of antiarrhythmic drugs on the cardiac sodium channel.

14-3-3's Effects on Quinidine Block. Quinidine is a class Ia AAD known to block $\text{Na}_v1.5$ in a voltage- and use-dependent manner. In the current study, we found that inhibiting 14-3-3 brought a more potent block by quinidine of the peak sodium current. With 20 μM quinidine, the peak sodium current was blocked by 21% in the control group and by 50% when 14-3-3 was inhibited. This result suggested that 14-3-3 inhibition might strengthen quinidine's effect on prolonging the cardiac action potential phase 0 (Salata and Wasserstrom, 1988). However, this enhancement is unlikely to be through modulation of the channel inactivation based on our observation that 14-3-3 inhibition removed the depolarizing shift caused by quinidine on SSI (Fig. 3A). Recent structural studies found that quinidine binds with the channel in the channel lumen as well as a fenestration between domain (D)-II and DIII. (Li et al., 2021). Since the fenestration is likely to be reshaped during channel gating and the accompanying conformational changes, it is plausible that 14-3-3's interaction with $\text{Na}_v1.5$ would affect this fenestration, thus changing quinidine's accessibility. Interestingly, sodium channel mutations near the

14-3-3 binding sites were also shown to alter the channel's response to quinidine (Itoh et al., 2005; Shurairh et al., 2007), further suggesting the possible involvement or influence of 14-3-3 in quinidine binding.

14-3-3's Influence on the Channel Activation. The protein 14-3-3 was first identified as a regulator of $\text{Na}_v1.5$ by yeast two-hybrid screening and functional studies in COS-7 cells (Allouis et al., 2006). The study also showed that $\text{Na}_v1.5$ activation was not influenced by 14-3-3 (Allouis et al., 2006). Here, in the present study, we focused on the impact of 14-3-3 on AADs' effects instead of the direct regulation of 14-3-3 on the channel. We found that 14-3-3 inhibition consistently shifted the channel's activation toward hyperpolarized voltages under the influence of all AADs studied. It is possible that the AADs' block could unmask the effects of 14-3-3 on channel activation. Alternatively, this might be explained by the fact that 14-3-3 might promote the channel activation under the influence of class I blockers. The influence of 14-3-3 inhibition on $\text{Na}_v1.5$ activation in the presence of AADs' is reminiscent of our previous observation that $\text{Na}_v1.5$ coupling presumably took a role in the channel activation (Clatot et al., 2017). The ubiquity of the 14-3-3 effects on channel activation under AADs' influences

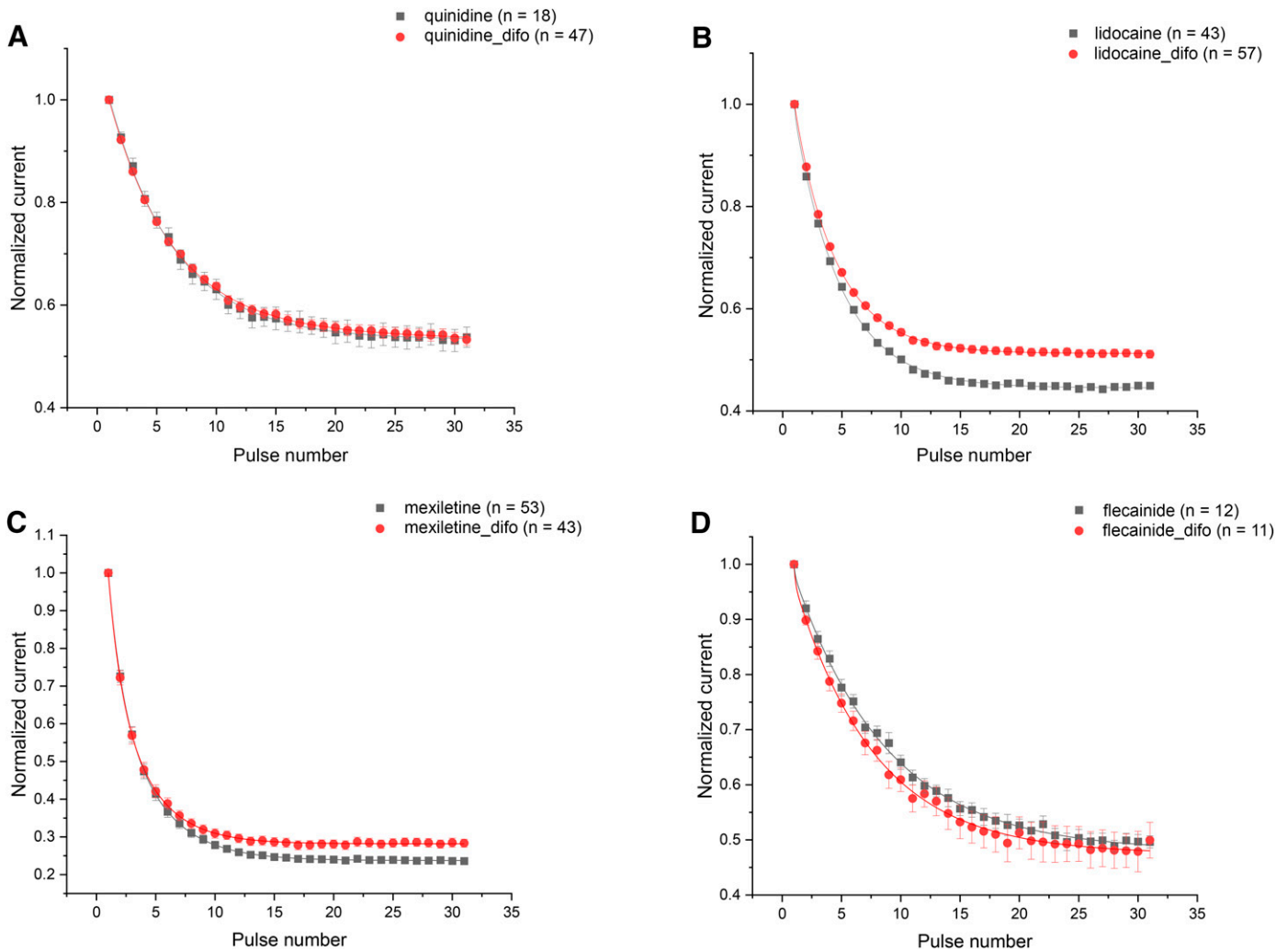


Fig. 8. Impact of 14-3-3 inhibition on the use-dependent block at 10 Hz stimulation. A train of 30 pulses was applied to the HEK293 cells. 14-3-3 inhibition diminished use-dependent block of lidocaine (B) and mexiletine (C) without affecting quinidine (A) and flecainide (D). difo, difopein.

indicated that the channel coupling could impact channel activation. On the other hand, these results also suggest that $\text{Na}_v1.5$ coupling most likely alters the channel's pharmaceutical response.

Among the four tested AADs, we found that class Ib AADs lidocaine and mexiletine shifted the activation curves toward depolarized voltages. Interestingly, the curves were shifted back toward the control with 14-3-3 inhibition (Fig. 2, B and C). This indicated that although class Ib AADs could delay $\text{Na}_v1.5$ activation, protein 14-3-3 is necessary for this regulation.

Regulation of Use-Dependent Block by 14-3-3. The voltage-gated sodium channel's responses to high-frequency stimulation can be blocked more readily than responses to low-frequency, known as use-dependent, blocking. This enhanced blocking potency during rapid stimulation can be explained by a varied equilibrium with the "reactivity" of the channel-drug interaction (Starmer et al., 1984). Based on our finding that 14-3-3 inhibition increased the acute quinidine block of the peak sodium current, one might expect to see effects of 14-3-3 inhibition on the quinidine's use-dependent block. However, use-dependent represents the accumulated effect of a drug during high-frequency repetitive stimulations. It reflects the rate of binding and release of the drug to and from the channel, which might not require the modified channel kinetics (Starmer et al.,

1984). It is therefore feasible to observe a more potent block of peak current but no effect on use-dependency. Indeed, we observed that the use-dependent block by quinidine was not affected by 14-3-3 inhibition (Fig. 9A).

We reported in this study that 14-3-3 inhibition significantly changed the use-dependent inhibition of class Ib AADs lidocaine and mexiletine. We observed that the use-dependent inhibition is not as significant for lidocaine and mexiletine at 1 Hz compared with other drugs (Fig. 6). This is consistent with previous findings and with the knowledge that class Ib AADs have relatively fast dissociation kinetics, thus being less affected by use-dependent block at low frequency (Wang et al., 2015; Shenasa et al., 2020). Here, in our study, the 14-3-3 inhibition had the tendency to increase the use-dependent block induced by lidocaine and mexiletine at 1 Hz. However, 14-3-3 impacted the use-dependent block of these two drugs at 5 and 10 Hz, impairing the use-dependent block (Fig. 9). This indicates that 14-3-3 could regulate $\text{Na}_v1.5$ and its interaction with AADs differentially with increased heart rate. Importantly, the effects on use-dependent block and the 14-3-3 inhibition showed similar patterns for the two class Ib drugs but different patterns for class Ia and Ic AADs, suggesting that the 14-3-3 regulation is in line with the mode of action of varied AAD subclasses.

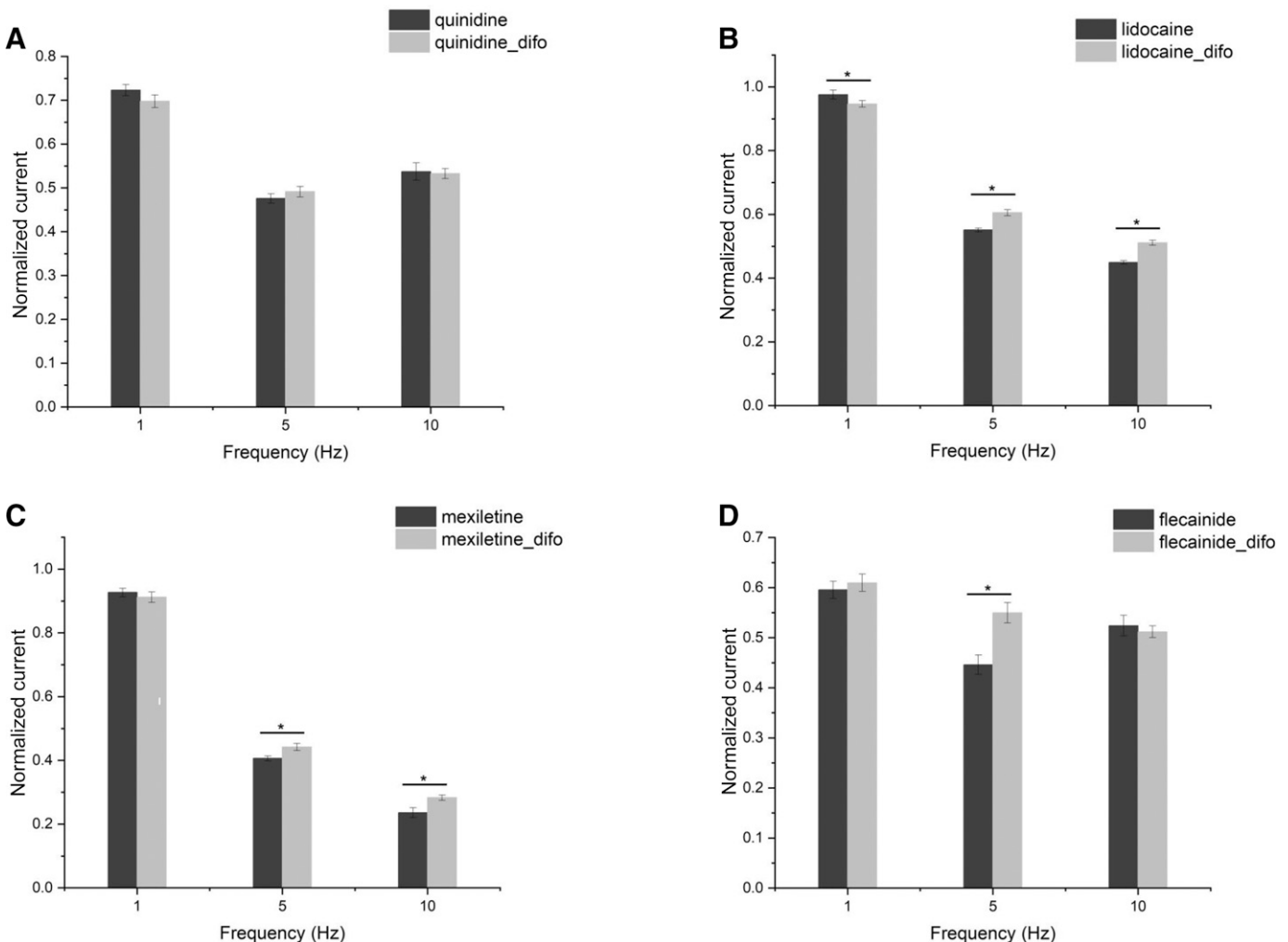


Fig. 9. Summary of the 14-3-3 effects on the use-dependent block of quinidine (A), lidocaine (B), mexiletine (C), and flecainide (D). difo, difopein.

Different Impact of 14-3-3 Regulation on Different AAD Subtypes. In the dose-dependent block experiments, we observed that 14-3-3 inhibition significantly lowered the IC₅₀ of quinidine, meaning that 14-3-3 increased Na_v1.5's sensitivity toward quinidine. Meanwhile, there was little effect on the dose-dependence of other tested drugs. This is consistent with the other results shown by the I/V curve, where 14-3-3 inhibition aggravated sodium current block by quinidine and the block by other drugs was not significantly changed (Fig. 1). In our experiments, the IC₅₀ of class Ic (flecainide), Ia (quinidine), and Ib (lidocaine and mexiletine) AADs were 84, 102, 198, and 286 μ M, respectively. These results reproduced the order of sodium channel block by these drugs in clinical practice and previous investigations, which was class Ic > Ia > Ib (Kecskemeti, 1991).

We also found that the 14-3-3 inhibition did not influence the dose-dependence of the class Ib AADs lidocaine and mexiletine. As discussed earlier, we observed a similar pattern for 14-3-3 inhibition effects on activation and use-dependent block with the two class Ib AADs. The two class Ib AADs acted similarly in response to 14-3-3 inhibition throughout our study, and this is consistent with AADs' mode of action and the molecular interaction between the channel and the drugs. Previous investigations demonstrated that only class Ib AADs bind with the channel via a strong electrostatic cation- π interaction, which could significantly influence the binding mode and affinity of a compound (Ragsdale et al., 1996; Liu et al., 2003; Pless et al., 2011; Tikhonov and Zhorov, 2017). It is possible that this interaction also led to the differential response of class Ib AADs to 14-3-3 inhibition. This specific and similar impact of 14-3-3 inhibition on the action of the two class Ib drugs might also suggest that the electrostatic cation- π interaction of the drugs with the channel is dependent on the biophysical coupling modulated by 14-3-3.

Structural and functional studies showed that the DIV S6 of Na_v1.5 is the key site for drug binding (Ragsdale et al., 1994; Ragsdale et al., 1996). However, the Na_v1.5 dimerization, as well as the 14-3-3 regulation, depend more on the DI-DII linker (Allouis et al., 2006; Clatot et al., 2017). Although further investigations on the three-dimensional structure are needed, the current understanding of the spatial relationship between the DI-DII linker and DIV S6 suggests that protein 14-3-3 is unlikely to influence the AAD's effects by direct interaction with the AAD's molecule or the binding sites. Our results showed that 14-3-3 inhibition has different effects on different drugs and different tendencies to promote or subdue the channel gating functions (i.e., activation, recovery, and SSI). This divergence further indicated that the 14-3-3 regulation of AADs' effect is likely not through 14-3-3 directly binding with drug, masking or revealing the binding sites. Instead, it is likely linked to the role that 14-3-3 plays in biophysical coupling of the sodium channel. 14-3-3 could also be involved through modulation of the sodium channel conformation in a steady state or changes in conformation during the channel gating. Thus, the DI-DII linker, due to the involvement of 14-3-3, could be another important site for the sodium channel's pharmaceutical effects due to its modulation of the biophysical coupling.

The results that difopein can diminish the use-dependent effect are particularly interesting as it is a direction that industry is looking for in drug development. Our findings suggest that a strategy targeting the specific inhibition of 14-3-3 action on the cardiac sodium channel could increase the efficacy of class

I antiarrhythmics. However, a broad inhibition of 14-3-3 is not a viable option in the clinical setting due to the promiscuous roles of 14-3-3 in human physiology. We believe that 14-3-3 has a fundamental role in cardiac electrophysiology and could be developed into a therapeutic target itself or shed light for expansion of novel therapeutics. Therefore, this study provided new information about how 14-3-3 influences the Na_v1.5 pharmaceutical responses and built the foundation for novel therapeutic strategies targeting 14-3-3 as it relates to arrhythmias.

In conclusion, we demonstrated that inhibiting protein 14-3-3 could greatly influence the cardiac sodium channel's response to AADs, including altered AAD block on the peak current, biophysical kinetics, and use- and dose-dependent block. Our observations reinforced the significance of Na_v1.5 coupling and indicated that this channel characteristic needs to be considered carefully in the study of Na_v1.5-related therapeutics. It also highlighted the necessity of future studies to reveal more structural information regarding the Na_v1.5 coupling and interacting proteins and how it relates to the action of AADs.

Authorship Contributions

Participated in research design: Zheng, Deschenes.

Conducted experiments: Zheng.

Performed data analysis: Zheng.

Wrote or contributed to the writing of the manuscript: Zheng, Deschenes.

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